WORKED SOLUTION

STEP I 2015, Question 3

Pure • Coordinate Geometry Visibility

STEP I 2015, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

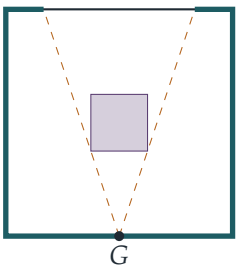
step.maths.org/questions/15-s1-q3

The key idea

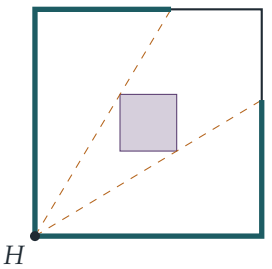
A guard's line of sight is cut off by the building, and since the building is a convex square, the lines that just graze one of its corners mark where the visible part of a wall ends. So for each position we draw these grazing lines, find where they meet the perimeter, and add up the visible lengths. Similar triangles (or the equation of a straight line) do all the work. The number $3a$ turns up as the value of b at which the grazing line from the middle of a wall reaches a corner of the courtyard.

Solution

Put the origin at the centre, so the perimeter wall is the square with sides on $x = \pm\frac{b}{2}$, $y = \pm\frac{b}{2}$, and the building is the square $|x| \leq \frac{a}{2}$, $|y| \leq \frac{a}{2}$, with $0 < a < b$. The total length of the perimeter is $4b$. A guard sees a point Q of the wall exactly when the segment from the guard to Q does not pass through the building.



middle of a wall ($b > 3a$)



corner

The thick (teal) parts of the wall are the ones the guard can see; the dashed lines graze corners of the building.

A guard at the middle of a wall

Take the guard at $G = (0, -\frac{b}{2})$, the middle of the bottom wall. The whole bottom wall is visible, since the building lies strictly above it: that is length b. By symmetry about the y-axis we only need to look to the right.

A line of sight going up and to the right misses the building exactly when it passes to the right of the corner $K = (\frac{a}{2}, -\frac{a}{2})$ nearest to it. (The line meets $y = -\frac{a}{2}$ first and its x-coordinate increases as it rises, so if it is to the right of the building at height $-\frac{a}{2}$, it stays to the right; if it is to the left of K there, it enters the building, because at height $-\frac{a}{2}$ its x-coordinate is between 0 and $\frac{a}{2}$.) So the boundary of what G sees is the line GK, which rises by $\frac{b-a}{2}$ over a run of $\frac{a}{2}$, so its gradient is $\frac{b-a}{a}$.

This line meets the right-hand wall $x = \frac{b}{2}$ at height

$$y_1 = -\frac{b}{2} + \frac{b}{2} \cdot \frac{b-a}{a} = -\frac{b}{2} + \frac{b(b-a)}{2a}.$$

It reaches the top of that wall ($y_1 \geq \frac{b}{2}$) exactly when $\frac{b-a}{a} \geq 2$, that is, when $b \geq 3a$.

? **Ask yourself** Why does the comparison of b with $3a$ come in here?

It decides which wall the grazing line GK hits: the right-hand wall if $b < 3a$, or the far (top) wall if $b > 3a$. The two cases give different formulas for the visible length.

Case $b < 3a$. The guard sees the right-hand wall from its bottom corner up to height y_1, a length

$$y_1 + \frac{b}{2} = \frac{b(b-a)}{2a},$$

and nothing beyond it (every line to a higher point of that wall, or to the top wall, passes to the left of K). With the matching piece on the left, the total visible length is

$$M = b + 2 \cdot \frac{b(b-a)}{2a} = b + \frac{b(b-a)}{a} = \frac{b^2}{a}.$$

Case $b > 3a$. Now the whole right-hand wall (length b) is visible, and the line GK meets the top wall $y = \frac{b}{2}$ after rising by b, so at

$$x = \frac{a}{b-a} \cdot b = \frac{ab}{b-a},$$

which is less than $\frac{b}{2}$ because $2a < b - a$. The guard sees the top wall from there to the corner, a length $\frac{b}{2} - \frac{ab}{b-a} = \frac{b(b-3a)}{2(b-a)}$. Adding the bottom wall, both side walls and both pieces of the top wall:

$$M = 3b + \frac{b(b-3a)}{b-a} = \frac{3b(b-a) + b(b-3a)}{b-a} = \frac{b(4b-6a)}{b-a}.$$

A guard in a corner

Take the guard at $H = (-\frac{b}{2}, -\frac{b}{2})$. The two walls through H are fully visible: length $2b$. The picture is symmetric about the diagonal $y = x$, so look at the right-hand wall. By the same argument as before, the visible part is cut off by the line through H and the corner $K = (\frac{a}{2}, -\frac{a}{2})$, which has gradient

$$\frac{\frac{b}{2} - \frac{a}{2}}{\frac{b}{2} + \frac{a}{2}} = \frac{b-a}{b+a}.$$

Over the run of b to the wall $x = \frac{b}{2}$ it rises by $\frac{b(b-a)}{a+b}$, which is less than b, so this line always hits the right-hand wall, whatever the values of a and b. The guard sees a length $\frac{b(b-a)}{a+b}$ of the right-hand wall, the same of the top wall, and none of the rest of either. Hence

$$C = 2b + \frac{2b(b-a)}{a+b} = \frac{2b(a+b) + 2b(b-a)}{a+b} = \frac{4b^2}{a+b}.$$

Comparing the two positions

Case $b < 3a$.

$$C - M = \frac{4b^2}{a+b} - \frac{b^2}{a} = \frac{b^2(4a - (a+b))}{a(a+b)} = \frac{b^2(3a-b)}{a(a+b)} > 0,$$

so the **corner** is preferable.

Case $b > 3a$.

$$M - C = \frac{b(4b-6a)}{b-a} - \frac{4b^2}{a+b} = \frac{b[(4b-6a)(a+b) - 4b(b-a)]}{(b-a)(a+b)} = \frac{2ab(b-3a)}{(b-a)(a+b)} > 0,$$

since $(4b-6a)(a+b) - 4b(b-a) = 4b^2 - 2ab - 6a^2 - 4b^2 + 4ab = 2ab - 6a^2$. So the **middle of a wall** is preferable.

When $b = 3a$ both formulas for M give $9a$, and $C = \frac{36a^2}{4a} = 9a$ too, so the two positions are equally good.

Checks

- Take $a = 1$, $b = 2$: $M = 4$ and $C = \frac{16}{3} \approx 5.33$, so the corner wins. With $a = 1$, $b = 5$: $M = \frac{5 \cdot 14}{4} = 17.5$ and $C = \frac{100}{6} \approx 16.67$, so the middle wins. These values were confirmed by a computer check that tested the line of sight to thousands of points around the wall.
- If $b \to a$ the building fills the courtyard: $M \to a$ (only the guard's own wall) and $C \to 2a$ (the two walls meeting at the corner), as they should.
- As $b \to \infty$ the building becomes negligible: $M = \frac{b(4b-6a)}{b-a}$ and $C = \frac{4b^2}{a+b}$ both approach the full perimeter $4b$, and both stay below it.

Common slips

- Forgetting that for $b > 3a$ the guard at the middle of a wall can see part of the opposite wall, not only the two side walls.
- Grazing the wrong corner of the building: from below, the corners that matter are the two bottom ones, $(\pm\frac{a}{2}, -\frac{a}{2})$.
- Leaving out the guard's own wall (or walls) from the total.

Beyond STEP

Questions about which parts of a region can be seen from a point belong to computational geometry: the classic "art gallery theorem" says that $\lfloor n/3 \rfloor$ guards always suffice to watch every point of a polygonal gallery with n walls, and sometimes that many are needed. Working out a visibility region, as we did here, is the basic step in such problems, and in computer graphics, where hidden-surface removal works the same way. For more problems of this style, Siklos, *Advanced Problems in Mathematics*, is a good next step.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.